

MFR Virtual Lab — Code Change Report

Mixed Flow Reactor (CSTR) teaching app · Flutter / Dart · Android

Commit	3179eb7573fedff8262c0ba1c3db63d3eee914fb
Message	fixed mfr_solver and excel_exporter
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Scope	7 files changed · +170 / -49 lines

1. Executive Summary

This change set is best understood as **one root-cause fix and its downstream consequences**. The graphical method the app uses to let a student determine the reaction rate constant k was based on a Y variable that was missing the inlet concentration term CA_0 . As a result, the best-fit line's slope equalled $k \cdot CA_0$ rather than k , so the value students read off the graph was wrong by a factor of CA_0 .

The commit:

- **Corrects the linearisation formula** in the solver so the graph slope is exactly k .
- **Adds four new physically-meaningful quantities** (CB , $CA \cdot CB$, r_A , k per trial) computed per trial, including a direct numerical cross-check of k .
- **Propagates** those quantities through the data model, provider, Excel sheet and on-screen graph labels.
- **Upgrades the Excel export** with a "Save to Device" vs "Share" choice and returns the saved file path to the UI.
- **Fixes an Android relaunch bug** when the app is reopened from the home-screen icon.

2. The Core Problem — the Math Fix

For a mixed flow reactor running the bimolecular reaction $A + B \rightarrow \text{products}$ with rate $-r_A = k \cdot CA \cdot CB$, the CSTR design equation rearranges into a straight line through the origin when plotted as Y vs τ :

$$Y = X_A / [C_{A0} \cdot (1 - X_A)(m - X_A)] = k \cdot \tau$$

where $m = CB_0 / CA_0$ and τ is space time. The slope of this line is the rate constant k .

Why it was a bug: the previous code computed $Y = X_A / [(1-X_A)(m-X_A)]$ — the CA_0 factor in the denominator was missing. That made $Y = k \cdot CA_0 \cdot \tau$, so the fitted slope (and every label, note and Excel column describing it) reported $k \cdot CA_0$ instead of k . The fix divides by CA_0 , so the slope is now k directly.

Before → After (mfr_solver.dart)

```
// BEFORE — slope came out as k·CA0
final double graphY = XA / ((1.0 - XA) * (m - XA));

// AFTER — divide by CA0 so the slope is k
final double graphY = XA / (CA0 * (1.0 - XA) * (m - XA));
```

New per-trial quantities (and the direct k cross-check)

```
final double CB      = CB0 - CA0 * XA; // unreacted B concentration [mol/L]
final double CACB    = CA * CB;       // product CA·CB [mol²/L²]
final double rA      = CA0 * XA / tau; // reaction rate of A [mol/L·min]
final double kPerTrial = rA / CACB;   // k = rA / (CA·CB) [L/mol·min]
```

Why add these: `kPerTrial` computes the rate constant *analytically* for each individual trial ($k = r_A / (CA \cdot CB)$). This gives an independent numerical value of `k` that students can compare against the slope obtained graphically — a built-in consistency check that also confirms the formula fix is correct. `CB`, `CA·CB` and `rA` are the intermediate quantities needed to show the full working in the exported spreadsheet.

3. File-by-File Breakdown

lib/utils/mfr_solver.dart CORE FIX

- Corrected the `graphy` linearisation formula (added the missing `CA0`).
- Added `CB`, `CACB`, `rA`, `kPerTrial` to the computation and to the `MFRResult` result object.
- Rewrote the explanatory comment block to describe slope = `k`; minor formatting tidy of the exception `toString()`.

Why: this is the source of truth for all reactor calculations. Fixing the formula here is what actually corrects the result; everything else in the commit exists to carry these new values to the user.

lib/models/trial_model.dart DATA MODEL

- Added four fields — `CB`, `CACB`, `rA`, `kPerTrial` — to `TrialModel` and its constructor.
- Included them in the row list, the display map, `toString()`, and the `kTrialExcelHeaders` list with proper units.

Why: `TrialModel` is the record stored per trial and consumed by the UI and the exporter. The new solver outputs have to live on the model or they can't reach the screen or the spreadsheet.

lib/providers/experiment_provider.dart DATA FLOW

- When building a `TrialModel` from an `MFRResult`, now copies across `CB`, `CACB`, `rA`, `kPerTrial`.

Why: the provider is the bridge between the solver's result and the stored trial. Without these four assignments the new fields would always be empty.

lib/utils/excel_exporter.dart EXPORT FIX

- Sheet 1 grew from 13 to 17 columns — added `CB`, `CA·CB`, `rA`, `k per trial`; column widths updated to match.
- Header and footer note text corrected: the Y formula now shows the `CA0` term and the note reads `slope = k` (was `slope = k × CA0`).
- `export()` / `exportToExcel()` now return `Future<String?>` (the saved path) instead of `Future<void>`.
- New `saveMode` parameter: `'save'` writes directly to the Downloads folder (Android `/storage/emulated/0/Download`, desktop `getDownloadsDirectory()`); `'share'` keeps the existing OS share-sheet flow.

Why: the spreadsheet is the student deliverable, so it had to show the corrected formula and the new working columns. Returning the file path and adding a direct-save option fixes the previous UX where users were only offered the share

sheet and never told where the file went.

lib/screens/home_screen.dart UI / UX

- Before exporting, shows a dialog: **Save to Device** or **Share File** (cancellable).
- Passes the chosen `saveMode` to the exporter and shows the returned save path in the confirmation snackbar (wrapped in `Expanded` so long paths don't overflow).

Why: surfaces the new exporter capability to the user and gives concrete feedback ("File saved to: ...") instead of a generic success message.

lib/widgets/mfr_graph.dart UI LABELS

- Y-axis title, slope readout, legend and doc-comments updated from `k·CA0` to `k`, and the Y formula now includes the `CA0` term.

Why: keeps the on-screen graph consistent with the corrected math — the displayed slope now genuinely is `k`, so the labels must say so.

android/.../MainActivity.kt ANDROID PLATFORM

- Override `onCreate`: if the activity is launched from the launcher icon and is **not** the task root, it calls `finish()` and returns instead of starting a second instance.

Why: fixes the common Android bug where tapping the home-screen icon while the app is already open spawns a duplicate activity / restarts the app, losing in-progress trial state.

4. New Spreadsheet / Trial Columns

Quantity	Symbol	Formula	Units
Concentration of B remaining	CB	$CB_0 - CA_0 \cdot XA$	mol/L
Concentration product	CA·CB	$CA \times CB$	mol ² /L ²
Rate of reaction of A	rA	$CA_0 \cdot XA / \tau$	mol/L·min
Rate constant (per trial)	k	$rA / (CA \cdot CB)$	L/mol·min
Graph Y variable (corrected)	Y	$XA / [CA_0(1 - XA)(m - XA)]$	—

5. Net Effect

- **Correctness:** the graphically-determined `k` now matches the true rate constant; the per-trial `kPerTrial` column provides an independent confirmation.
- **Transparency:** the exported workbook now shows the full intermediate working (CB, CA·CB, rA, k) instead of jumping straight to the plot variable.
- **Usability:** users can save the file directly to Downloads and see exactly where it landed.
- **Stability:** Android no longer relaunches/duplicates the app from the icon.

